Applied Machine Learning – Final Project

ECG Heartbeat Image Classification

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Team Responsibilities

At the beginning of the project, each member independently implemented one preprocessing pipeline, two supervised models, and two unsupervised models. After comparing performance results, we collaboratively selected the best-performing approaches for further analysis. During report writing, each member contributed by documenting and refining the sections corresponding to their selected models, including implementation details, performance discussion, and references. The final report was jointly edited and reviewed by the entire team to ensure consistency in presentation and analysis.

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Abstract

In this project, we classify ECG heartbeat images into five categories using computer vision techniques. The dataset is highly imbalanced, with most samples belonging to the normal class, and the test set includes distribution shift to simulate data from different hospitals or devices. We explored multiple preprocessing strategies, data augmentation methods, and supervised deep learning models based on ResNet, DenseNet, and EfficientNet architectures. Each group member independently developed a final model with different design choices.

Through our experiments, we observed that individual models are conservative and tend to over-predict the normal class due to severe class imbalance. To address this issue, we designed an ensemble decision rule that combines predictions from three models to reduce false normal predictions while maintaining stable detection of abnormal classes. Individually, the models achieved public macro-F1 scores of 0.72812 (DenseNet), 0.73530 (ResNet-18 with attention), and 0.74714 (EfficientNet). Our final ensemble reached a macro-F1 score of 0.76374, demonstrating improved robustness and generalization compared to any single model.

1. Introduction

Electrocardiograms (ECGs) are widely used to diagnose heart diseases. In many real-world scenarios, ECG signals are only available as images, such as scanned medical records or data from legacy systems. Therefore, reliable ECG image classification is important for automated and large-scale diagnosis.

In this project, ECG time-series signals are converted into 128×128 grayscale images, and the task is to classify each image into one of five heartbeat categories. The training dataset contains 87,554 labeled images, while the test dataset contains 21,892 unlabeled images. A major challenge is extreme class imbalance: over 80% of the training data belongs to class 0 (normal), as shown in Figure 1. In addition, the test data distribution is shifted to simulate ECGs collected from different hospitals or devices.

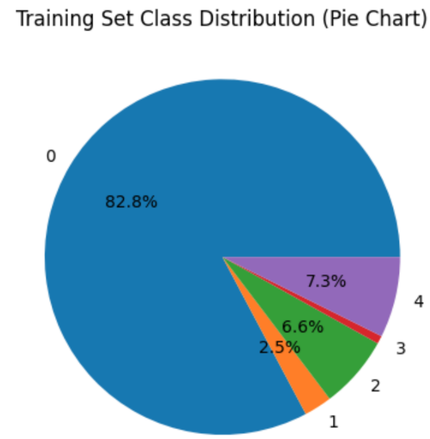

Figure 1. Training set class distribution. Class 0 (normal) dominates the dataset, accounting for more than 80% of all samples.

Figure 2 shows example training images from class 3, illustrating large intra-class variation in waveform shape and noise patterns. These challenges motivate the use of strong data augmentation, imbalance-aware training strategies, and ensemble methods. Our goal is to build models that generalize well under severe class imbalance and distribution shift.

2. Related Work

Previous ECG classification work mainly focuses on time-series models applied to raw ECG signals. Common approaches include convolutional neural networks (CNNs) to capture local waveform patterns and recurrent neural networks (RNNs), such as LSTMs, to model temporal dependencies. Some methods combine CNNs and LSTMs to jointly learn spatial and sequential features [1].

More recent studies introduce attention mechanisms to improve performance by allowing models to focus on important waveform segments, which is especially helpful for minority classes [1]. Ensemble methods are also widely used in medical diagnosis to improve robustness and stability under class imbalance. To address imbalance, prior work explores techniques such as synthetic data generation

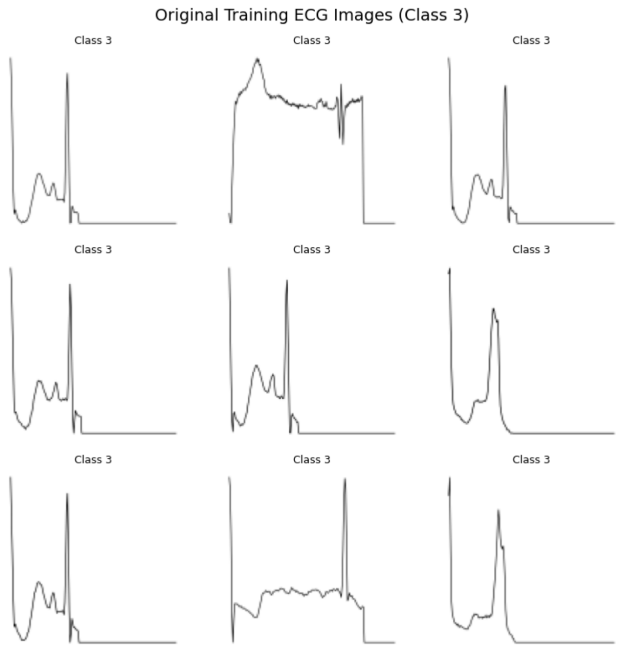

Figure 2. Example ECG training images from class 3. The samples show large variation in waveform shape, amplitude, and noise, highlighting the difficulty of classification.

and class-balanced sampling [2]. Overall, these studies suggest that combining different model architectures, attention mechanisms, and ensemble strategies is effective for ECG classification under class imbalance and distribution shift [3].

3. Methods

Our group first built a set of CNN baselines to understand the dataset difficulty and to identify the main bottlenecks (class imbalance and distribution shift). After that, each member developed a stronger model with a different focus (architecture, augmentation, or domain generalization). Finally, we combined the three models with a simple rule-based ensemble to reduce "false normal" predictions.

3.1. Early Exploration: Baseline CNN Depth Study

Before using large pretrained networks, we tested small CNNs with different depths (2-layer, 4-layer, 6-layer, and 8-layer) using the same basic training pipeline (resize, normalize, cross-entropy, Adam). Our motivation was to check whether simply increasing capacity can solve the problem.

On the Kaggle public leaderboard, the macro-F1 improved quickly from 2-layer to 6-layer, but then saturated and even decreased for deeper models. For example, the 8-layer baseline achieved only 0.63468 on the public leaderboard in our runs, suggesting that deeper "from-scratch" CNNs do not automatically generalize better under distribution shift. We also tried adding Squeeze-and-Excitation (SE) attention on top of the 6-layer CNN, which only produced a small improvement (reported as 0.67647).

Baseline Model	Public Macro-F1	Observation
2-layer CNN	0.33020	Underfits; weak features
4-layer CNN	0.52409	Better, still limited
6-layer CNN	0.67472	Best among plain CNNs
8-layer CNN	0.63468	Worse generalization

Table 1. Baseline CNN depth exploration (public leaderboard). The 8-layer performance is consistent with our recorded result.

Although training F1 could reach around 0.9889 for 8 layers CNN, the leaderboard macro-F1 stayed around ~ 0.67 for these baselines. This gap indicates that overfitting and distribution shift are major issues, and we need (1) better feature extractors and (2) better robustness techniques. This is why our final approaches relied on pretrained backbones, stronger augmentation, and explicit imbalance / domain-shift handling.

3.2. Data Preprocessing

All approaches treat each ECG as a single-channel (grayscale) image. The common preprocessing steps are:

- **Resize:** images are resized to match the backbone input size (e.g., 224×224 for EfficientNet).
- **Tensor conversion:** convert PNG into a normalized tensor in $[0, 1]$.
- **Normalization:** we used a simple grayscale normalization (mean 0.5, std 0.5) as a stable default.

Some models also applied ECG-specific preprocessing to make waveforms clearer under noisy or shifted styles:

- **Line thickening and inversion:** make faint waveforms more visible and reduce sensitivity to background style.

3.3. Data Augmentation

Because the test set includes a synthetic distribution shift, augmentation is important for generalization. Across our experiments, we used:

- **Geometric jitter:** small rotations / translations to simulate different scan alignment.
- **Brightness / contrast changes:** simulate different devices and imaging conditions.
- **Style randomization:** random sharpness and color/contrast jitter to mimic hospital/device differences.

- **Mixup:** interpolate samples and labels to reduce overconfidence (used by Andrew).

In addition, Andrew applied test-time augmentation (TTA): run inference multiple times with small random transforms and average the predictions. In his implementation, TTA is performed 5 times, including a small affine translation and the thickening/inversion transform.

3.4. Individual Final Models (Breaking the ~ 0.67 Baseline Ceiling)

From our baseline experiments, we learned that training a CNN from scratch is not enough for this task. Even though deeper CNNs perform very well on the training set, their performance on the leaderboard stops improving around a macro-F1 score of 0.67. This happens mainly because the dataset is highly imbalanced and the test data comes from a different distribution. To overcome this limitation, our final models focus on three main ideas:

1. Using strong pretrained models to extract better features
2. Improving robustness with data augmentation and test-time augmentation
3. Addressing the imbalance and shift using sampling, focal loss, and domain regularization

3.4.1 David: ResNet-18 with ECG-Specific Augmentation and Spatial Self-Attention (Public Macro-F1 = 0.73530)

David explored a sequence of models based on ResNet architectures to improve generalization under severe class imbalance and distribution shift [4]. His first model used a standard ResNet backbone with grayscale ECG images and achieved limited performance (public macro-F1 of 0.67787). This result showed that a plain pretrained CNN, while stronger than shallow baselines, still struggles to generalize well to the shifted test distribution.

To improve robustness, David designed ECG-specific data augmentation that simulates realistic variations in heartbeat signals. These augmentations include amplitude scaling, baseline drift, Gaussian noise, and temporal warping. Together, they model changes caused by different devices, noise levels, and recording conditions. Visualization of augmented samples shows that waveform shape is preserved while meaningful variability is introduced.

Building on this augmentation strategy, David introduced a spatial self-attention module on top of ResNet-18 [5]. After the final convolutional layer, the feature map is flattened and passed through a multi-head self-attention block. This allows each spatial location to attend to other locations in the image, enabling the model to capture long-range relationships across the ECG waveform. The attention output is combined with the original features using a residual connection and then fed into the classifier. This final model achieved a public macro-F1 score of 0.73530.

Why this helped. ECG-specific augmentation exposes the model to realistic signal variations, improving robustness to distribution shift. Spatial self-attention enables the model to focus on important waveform regions and capture global dependencies that standard convolutions may miss. Together, these design choices improve feature representation and generalization without simply increasing model depth.

3.4.2 Andrew: DenseNet-121 with ECG-Specific Augmentation and Test-Time Augmentation (Public Macro-F1 = 0.72812)

Andrew focused on improving robustness to noise, image style variation, and class imbalance. His final model uses DenseNet-121 as the backbone and processes ECG images as single-channel grayscale inputs [6]. This design preserves the original signal structure and avoids introducing unnecessary color information.

During training, images are resized to 224×224 and augmented with small rotations and translations, brightness and contrast changes, waveform thickening, color inversion, and Mixup [7]. Among these, the line-thickening and inversion operation plays an important role by highlighting ECG waveforms and reducing background noise. To further address class imbalance, Andrew applied a class-weighted loss that assigns higher importance to rare heartbeat classes.

At inference time, Andrew applied test-time augmentation [8]. Each test image is evaluated five times using small affine transformations, and the final prediction is obtained by averaging the outputs. This reduces prediction variance and improves stability when the test data distribution differs from the training data.

Andrew also experimented with a modified version of the model, which converted grayscale images to RGB, increased the batch size, and used ImageNet normalization. However, this variant did not improve performance. As a result, the grayscale-based Model achieved the best performance and was selected as Andrew's final model.

Why this helped. The ECG-specific preprocessing emphasizes waveform structures that are important for classification while reducing sensitivity to background noise. Strong data augmentation and Mixup reduce overfitting and improve robustness to distribution shift. In addition, test-time augmentation stabilizes predictions by averaging multiple views of each sample, leading to more reliable performance on unseen data.

3.4.3 Alice: EfficientNet with Imbalance-Aware Training and Domain Regularization (Public Macro-F1 = 0.74714)

Alice focused on directly addressing the two main challenges of the dataset: severe class imbalance and distribution shift between training and test data. She used EfficientNet-B0 as the backbone, which provides strong feature extraction while remaining computationally efficient [9]. All ECG images are treated as single-channel grayscale inputs and resized to 224×224 to match the network input size.

To handle class imbalance, Alice applied Weighted Random Sampling so that minority classes are sampled more frequently during training, a common strategy for learning from imbalanced data [10]. She also adopted focal loss, which reduces the influence of easy samples and encourages the model to focus on harder, often minority-class examples [11]. This prevents the training process from being dominated by the normal class.

To improve robustness to distribution shift, Alice introduced style randomization during training. These transformations include changes in brightness, contrast, sharpness, small rotations, and Gaussian blur, simulating variations caused by different hospitals, devices, and recording conditions. In addition, she incorporated a domain regularization term based on Maximum Mean Discrepancy (MMD), which encourages the feature distributions of original and style-shifted images to be similar [12].

The final training objective combines focal loss for classification and MMD loss for domain alignment. This design encourages the model to learn discriminative yet domain-invariant features. With careful hyperparameter tuning and early stopping, Alice's model achieved the strongest single-model performance with a public macro-F1 score of 0.74714.

Why this helped. Weighted sampling and focal loss improve learning on rare heartbeat classes and reduce bias toward the normal class. Style randomization exposes the model to diverse visual conditions, while MMD regularization encourages stable feature representations across different domains. Together, these components significantly improve generalization under distribution shift.

3.5. Ensemble Strategy

From validation and submission results, we observed that all individual models are conservative due to the dominance of the normal class. As a result, models tend to predict class 0 unless there is strong evidence of abnormality, which can hurt macro-F1 by missing rare classes.

To address this issue, we designed a simple rule-based ensemble instead of standard majority voting or probability averaging. Since false normal predictions are more harmful under macro-F1, the ensemble is biased toward detecting abnormality whenever at least one model provides confident evidence.

The ensemble decision rules are as follows:

- If all three models predict class 0, the final prediction is class 0.
- If any model predicts a non-zero class, the sample is treated as abnormal.
- If only one model predicts an abnormal class (1–4), that prediction is used.
- If multiple models predict different abnormal classes, conflicts are resolved using a fixed priority order: Alice $\rightarrow$ David $\rightarrow$ Andrew.

The priority order is determined by each model's public leaderboard macro-F1 score. Alice's model achieved the highest macro-F1 among the three, followed by David's and then Andrew's. This ranking reflects overall generalization performance and provides a principled way to resolve conflicting abnormal predictions. The resulting ensemble reduces false normal predictions while remaining interpretable and stable.

4. Results

We evaluate all models using the macro-F1 score on the Kaggle public leaderboard, which treats all classes equally and is therefore suitable for highly imbalanced datasets. We report results for individual models as well as the final ensemble.

4.1. Individual Model Performance

Table 2 summarizes the performance of each group member's final model. Among single models, Alice's EfficientNet-based approach achieved the highest macro-F1 score, followed by David's attention-enhanced ResNet and Andrew's DenseNet with heavy augmentation.

Model	Public Macro-F1
Andrew (DenseNet-121)	0.72812
David (ResNet-18 + Attention)	0.73530
Alice (EfficientNet + Imbalance Handling)	0.74714

Table 2. Public leaderboard performance of individual models.

Although all three models significantly outperform baseline CNNs, their prediction behaviors differ. Andrew's model shows strong robustness to noise and style variation, David's model captures global waveform relationships through self-attention, and Alice's model explicitly handles class imbalance and domain shift. These complementary strengths motivate the use of an ensemble.

4.2. Ensemble Performance

The final rule-based ensemble combines the predictions of all three models. The ensemble achieves a macro-F1 score of **0.76374**, outperforming all individual models.

Method	**Public Macro-F1**
Best Single Model (Alice)	0.74714
Final Ensemble	**0.76374**

Table 3. Comparison between the best single model and the final ensemble.

The improvement demonstrates that the ensemble effectively reduces false normal predictions, which is critical under macro-F1 evaluation. By flagging a sample as abnormal whenever at least one strong model predicts an abnormal class, the ensemble improves recall on minority classes while remaining conservative for the normal class.

4.3. Analysis and Insights

Several important insights emerge from the results. First, simply increasing model depth or capacity is not sufficient; baseline CNNs saturate around a macro-F1 of 0.67. Second, explicitly addressing dataset challenges such as class imbalance and distribution shift leads to consistent performance gains. Models that incorporate imbalance-aware loss functions, domain generalization techniques, or robustness-focused augmentation outperform those that do not.

Finally, the ensemble strategy provides an additional boost by combining complementary model behaviors. Instead of relying on majority voting, the rule-based design leverages model confidence asymmetrically, which aligns well with the macro-F1 objective. Overall, the results show that careful model design and ensemble reasoning are both essential for strong performance on this task.

5. Discussion

In this project, we studied ECG heartbeat image classification under severe class imbalance and distribution shift. Through extensive experimentation, we found that simply increasing model complexity is not sufficient to achieve strong generalization. Instead, performance improvements come from addressing the specific challenges of the dataset, including imbalance, noise, and domain variation.

One key observation is that baseline CNNs trained from scratch quickly overfit the dominant normal class and saturate around a macro-F1 score of 0.67. Even deeper CNNs fail to generalize better without additional mechanisms. This motivated the use of pretrained backbones, which provide stronger feature representations and faster convergence. Among the individual models, approaches that explicitly handle class imbalance and distribution shift consistently achieved higher macro-F1 scores.

Each final model contributes a different strength. David's model improves feature representation by introducing spatial self-attention, which captures global waveform relationships. Andrew's model emphasizes robustness through ECG-specific augmentation and test-time augmentation, reducing sensitivity to noise and style changes. Alice's model directly addresses the core dataset challenges using weighted sampling, focal loss, and domain regularization. These complementary behaviors explain why the ensemble outperforms all single models.

The rule-based ensemble plays an important role in improving final performance. Since all individual models are conservative and biased toward predicting the normal class, standard majority voting is not ideal. Our ensemble strategy prioritizes abnormal detection whenever at least one strong model predicts an abnormal class, which aligns well with the macro-F1 evaluation metric. This design reduces false normal predictions while remaining simple and interpretable.

Despite these successes, our approach has several limitations. First, the ensemble relies on a fixed priority order based on public leaderboard performance, which may not be optimal for all datasets. Second, we did not explicitly calibrate model confidence scores, which could further improve ensemble decision-making. Finally, computational constraints limited the exploration of larger architectures or more extensive hyperparameter tuning.

There are several directions for future work. One possibility is to use confidence-aware or uncertainty-based ensemble methods instead of fixed rules. Another direction is to explore self-supervised or contrastive pretraining directly on ECG images to reduce dependence on ImageNet features. In addition, more advanced domain generalization techniques or adaptive thresholding strategies could further improve performance under distribution shift.

Overall, this project highlights the importance of combining model design, data-centric strategies, and task-aware ensembling. By carefully analyzing dataset challenges and aligning modeling choices with the evaluation metric, we are able to significantly improve performance beyond baseline approaches.

References

[1] Sun, G., Li, S., Zhang, J., and Li, Q. Automatic cardiac arrhythmias classification using CNN and attention-based RNN network. *Healthcare Technology Letters*, 10(2):53–61, 2023.

[2] Cheng, J., Zou, Q., and Zhao, Y. ECG signal classification based on deep CNN and BiLSTM. *BMC Medical Informatics and Decision Making*, 24(1):1–11, 2024.

[3] Kaggle Notebook: ECG Heartbeat Classification with CNN, LSTM, Attention, and Ensemble Methods.

[4] He, K., Zhang, X., Ren, S., Sun, J. Deep residual learning for image recognition *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.

[5] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 5998–6008, 2017.

[6] Huang, G., Liu, Z., Laurens, V. D. M., Weinberger, K. Q. Densely connected convolutional networks. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4700–4708, 2017.

[7] Zhang, H., Cisse, M., Dauphin, Y. N., Lopez-Paz, D. mixup: Beyond empirical risk minimization. *International Conference on Learning Representations (ICLR)*, 2018.

[8] Moshkov, N., Mathe, B., Kertesz-Farkas, A., Hollandi, R., Horvath, P. Test-time augmentation for deep learning-based cell segmentation on microscopy images. *Scientific Reports*, 10(1): Article 5068, 2020.

[9] Tan, M., and Le, Q. V. EfficientNet: Rethinking model scaling for convolutional neural networks. *International Conference on Machine Learning (ICML), Proceedings of Machine Learning Research (PMLR)*, 97: 6105–6114, 2019.

[10] He, H., and Garcia, E. A. Learning from Imbalanced Data. *IEEE Transactions on Knowledge and Data Engineering*, 21(9):1263–1284, 2009.

[11] Lin, T., Goyal, P., Girshick, R. B., He, K., and Dollár, P. Focal Loss for Dense Object Detection. In *IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988, 2017.

[12] Long, M., Cao, Y., Wang, J., and Jordan, M. I. Learning Transferable Features with Deep Adaptation Networks. *International Conference on Machine Learning (ICML)*, pages 97–106, 2015.